Quadratics: Introduction

Questions 1–23

Question 1. Algebraic Manipulation and Solving Equations Fully factorise the following:

a) $x^2 + 14x + 45$
b) $x^2 + 3x - 18$
c) $x^2 - 7x$
d) $2x^2 + 11x - 6$
e) $2x^2 - 17x + 35$
f) $8x^2 - 14x + 3$
g) $6x^2 + 17x + 5$
h) $-2x^2 - x + 3$
i) $x^2 - 9$
j) $4x^2 - 5$

Question 2. Fully factorise the following where a and b are constants:

a) $x^2 - a^2$
b) $(ax)^2 - b^2$
c) $x^2 - a$

Question 3. Fully factorise the following:

a) $8(x+2) - x(x+2)$
b) $-(x-2) + 2x(x-2)$
c) $12(x-5)^2 + (x-5)$
d) $(x-7)^2 - 4$
e) $\frac{2}{3}(3x-10)^2 - \frac{32}{3}$

Question 4. Solve the following:

a) $x^2 = x$
b) $x^2 - 4 = 0$
c) $x^2 - 8 = 0$
d) $x^2 + 2x = 0$
e) $(x-5)^2 = 0$
f) $x^2 - 12x + 36 = 0$
g) $\frac{1}{5}x^2 - \frac{1}{5}x = 4$
h) $(x+3)^2 = 6$
i) $(x-1)(x-3) = 8 - 2x$

Question 5. Solve the following, giving your answers in exact form:

a) $x^2 - 8x + 5 = 0$
b) $2x^2 - 40 = 0$
c) $2x^2 + 16x + 24 = 0$
d) $3x^2 - 5x - 1 = 0$
e) $ax^2 - b = 0,\ a \neq 0$
f) $ax^2 - bx = 0,\ a \neq 0$

Question 6. Two students are trying to find the solutions to $(x-5)^2 = (x-5)$. In each case, explain the mistake they made and then show your workings for correctly solving it.

Student 1:
When $x = 5$, $(5-5)^2 = (5-5)$
so $0 = 0$ which means $x = 5$

Student 2:
Divide both sides by $x - 5$: $x - 5 = 1$
so $x = 6$

Question 7. Solve $x - 3 = 2\sqrt{x}$.

Question 8. Solve $2x + 2 = 5\sqrt{x}$.

Question 9. By using the substitution $p = x^2$ find the solutions to $2x^4 - 11x^2 + 5 = 0$.

Question 10. By using the substitution $p = x^3$ find the solutions to $3x^3 + 19 = \frac{40}{x^3}$.

Question 11. By using the substitution $p = 2x^2 - x$, find the solutions to $12x^4 - 12x^3 + 5x^2 - x - 2 = 0$.

Completing the Square

Question 12. Complete the square on each of the following:

a) $y = x^2 - 8x$
b) $y = x^2 + 6x$
c) $y = x^2 - 10x + 40$
d) $y = x^2 + 8x - 20$
e) $y = x^2 + 5x + 7$
f) $y = 2x^2 - 16x - 20$
g) $y = 4x^2 - 24x + 25$
h) $y = 3x^2 - 7x + 1$

Question 13. Complete the square on each of the following:

a) $y = -x(x-2)$
b) $y = (x+3)(x+2)$
c) $y = -2x^2 - 4x + 5$
d) $y = -2x^2 - 7x + 1$
e) $y = -3x^2 + 5x + 10$
f) $y = 25 - 4x^2$
g) $y = x^2 + px + r$

Question 14. Explain the graphical relevance of a and b in the curve $y = (x-a)^2 + b$.

Question 15. For each of the following curves, find the coordinates of the vertex and state the type of vertex:

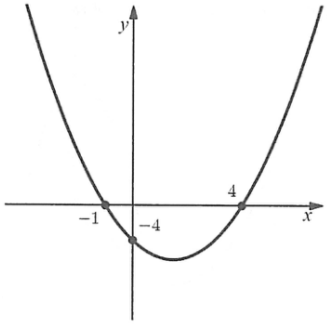
a) $y = (x-2)^2 - 12$
b) $y = (x+2)^2 - 8$
c) $y = -(x-1)^2 + 12$
d) $y = -(x+2)^2$
e) $y = 3(x+4)^2 + 5$
f) $y = 2(x+\frac{5}{2})^2 + \frac{1}{2}$
g) $y = -10(x-4)^2 - 100$
h) $y = x^2 - 6x - 2$
i) $y = 3x^2 + 6x + 5$
j) $y = -2x^2 - 4x + 3$
k) $y = 2x^2 - 5$
l) $y = -5x^2 + 12$
m) $y = a(x-b)^2,\ a < 0$
n) $y = x^2 + px + r$

Question 16. State the line of symmetry in the following curves:

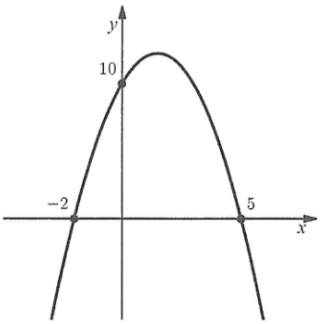
a) $y = (x-4)^2 + 5$
b) $y = 2(x+3)^2 + 7$
c) $y = -3(x-1)^2 - 6$
d) $y = (x-2)(x+4)$
e) $y = 2(x-1)(x-6)$
f) $y = 5(x+2)(x-8)$
g) $y = (x-a)(x-b)$
h) $y = (x-p)^2 + q$

Question 17. In each case, find the equation of the curve in the form $y = (x - a)(x - b)$ and state the coordinates of the vertex:

a)

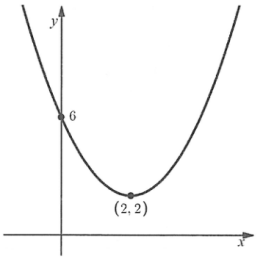


b)

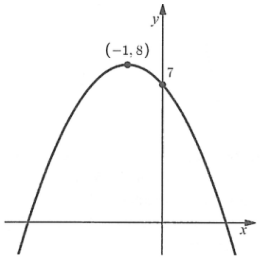


Question 18. For each of the following functions, find the equation of the graph in the form $y = a(x - b)^2 + c$ where a, b and c are values to be found:

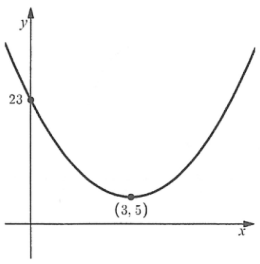
a)



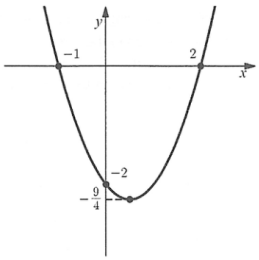
b)



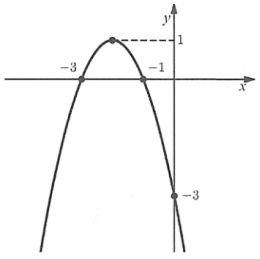
c)



d)



e)



Question 19. Find the equation of the curve in the form $y = ax^2 + bx + c$ for the curves described below:

a) Minimum point at $(1, 5)$, intersecting the y-axis at $y = 6$
b) Minimum point at $(3, -1)$, intersecting the y-axis at $y = 17$
c) Maximum point at $(-2, 4)$ intersecting the y-axis at $y = -76$

Question 20. Simultaneous Equations

We suggest that Algebra: Simultaneous Equations is helpful with Q20-Q23 Find the solutions to following pairs of simultaneous equations:

a) $y = 2x + 3$, $y = x^2 + x + 1$
b) $y = -2x + 3$, $y = x^2 - 2x + 2$
c) $y = 2 - x$, $y = 6 - x^2 - x$
d) $y = -2x^2 + x + 4$, $y = -x^2 + x + 3$
e) $y = x^2 - 2x$, $y = 2x^2 + x - 4$

Question 21. Find the point(s) of intersection of the following curves and straight lines:

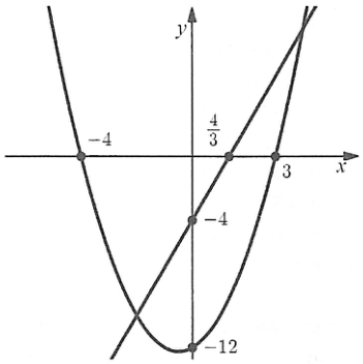
a) $y = x^2 + 1$, $y = \frac{1}{2}x + 1$

b) $y = -x^2 + 5$, $y = -x + 3$

c) $y = x^2 - 4x + 5$, $y = 2x - 4$

d) $y = 2x^2 - 4x + 3$, $y = 4x - 5$

Question 22. Find the points of intersection of the quadratic curve and the straight line below:



Question 23. Find the points of intersection of the quadratic curves shown in the graph below:

